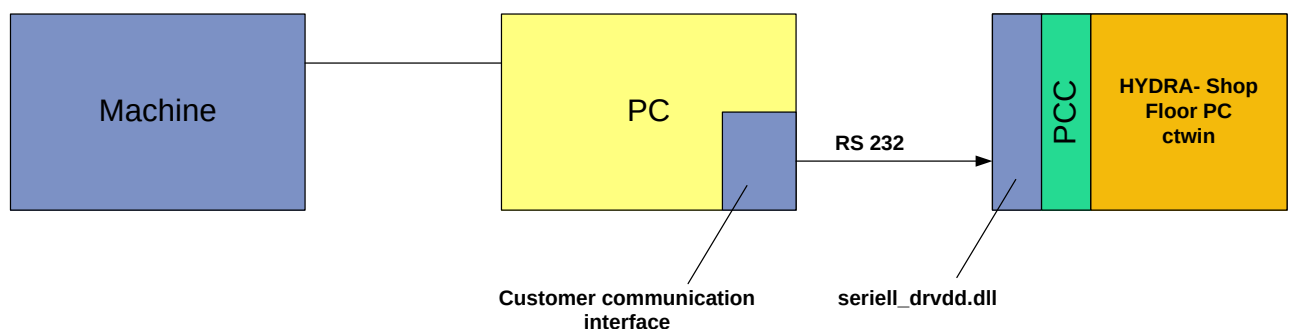


1 PCC Module Serial Interface

1.1 Machine Data Connection via RS232

Data transmission requires a serial RS-232 interface. It is possible to connect both the information of several machines via a serial channel as well as individual machines via one serial interface each.



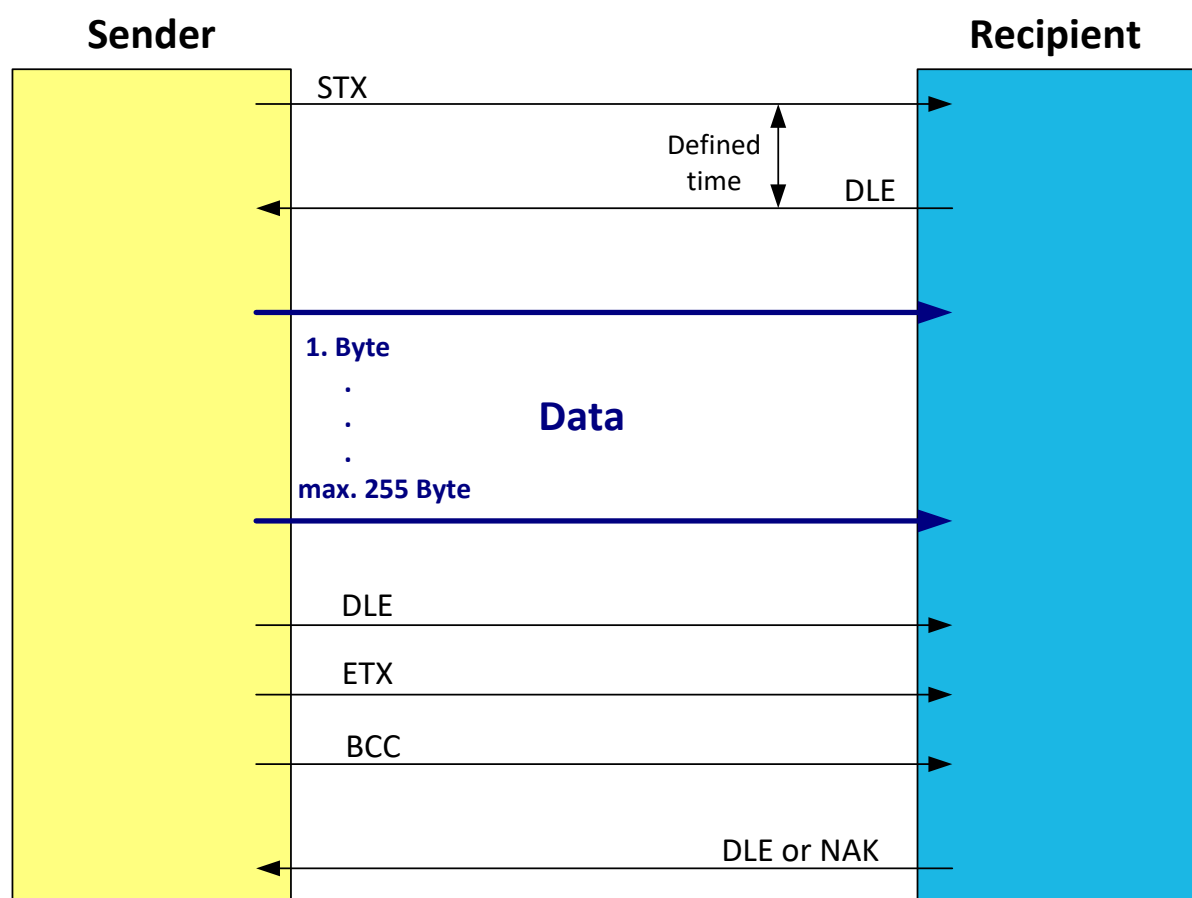
The data are transmitted in point-to-point operation. The control characters used are STX, DLE, ETX, NAK and BCC. The data block length must not exceed 255 bytes. The maximum block length is only applicable to net data, hence the data exclusive of control characters.

1.1.1 Explanation of Control Characters

Control character	Hexadecimal encoding	Explanation
STX	02	Sending STX signals sending readiness to the partner. Confirmation is expected within a predefined period.
ETX	03	DLE ETX signals the end of a transmission block
DLE	10	DLE signals readiness for receipt after receipt of STX. DLE is the positive confirmation of a correctly received transmission block. In addition, DLE initiates the end control character ETX. If DLE appears in the data block by chance, the driver adds a second DLE in order to make the recipient aware of this peculiarity. The receiver station will hide the additional DLE again.
NAK	15	NAK is the negative confirmation if a block was received with errors.

BCC		The block check character is a block-specific check sequence which is added to the data block and is used for recognizing incorrectly transmitted data and control blocks in the transmission of code-based character strings. During formation, error protection procedures are applied in accordance with the determination of parity bits. The block check character is the even longitudinal parity (EXOR link of all data bytes) of a sent and/or received block.
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1.1.2 Log Structure for Data Transmission



Data packages can be exchanged in both directions. For this purpose, the machine control provides information about the current machine conditions and in turn receives messages for signal lamps or machine locks from the HYDRA system in accordance with the configuration.

Data exchange follows the pattern described below:

- The sending system transfers a control character <STX> and signals that data are to be sent to the destination device.

- If the destination device is ready for receipt, it signals this by sending a control character <DLE>. If this confirmation is not received, the sending system starts two more attempts with the start character <STX> before it ceases sending, and then transmits an appropriate message to the higher-level system.
- In the case of on-time <DLE> confirmation by the destination device, the sender now starts transmitting the data package and adds <ETX> and <BCC> at the end of a sequence of <DLE>. After sending these control characters, the sending system awaits a reply from the destination device in order to complete the transmission of this data package.
- If a data package was received from the destination device, the latter will start evaluating the block check character (BCC) and hence determines the correct data contents of the block received. If no error was detected, a confirmation with control character <DLE> is sent. Otherwise, an error message <NAK> is sent to inform the sender that the package has to be repeated. This repetition, however, will start from step 1 with an <STX> control character.
- A <DLE> confirmation signals successful completion of the transmission to the sender. Subsequently, other transmissions may follow, or data may be exchanged in the reverse direction.